MUBEEN AFZAL

AI Engineer | Cybersecurity Researcher | Multi-Agent Systems Developer

Saarbrücken, Germany +49 152 14055115 mubeen.afzal2001@gmail.com
mubeen.live linkedin.com/in/mubeen-afzal-ai github.com/mubeen-afzal

Research Interests

AI-driven cybersecurity, adversarial machine learning, vulnerability assessment, privacy-preserving AI, secure multi-agent systems, trustworthy AI, and threat intelligence automation.

Education

M.Sc. Cybersecurity **Oct 2025 – Aug 2027**
Saarland University, Saarbrücken, Germany

B.Sc. Artificial Intelligence **Jul 2019 – Jan 2024**
National University of Computer & Emerging Sciences (FAST-NUCES), Islamabad, Pakistan

Technical Skills

Programming: Python, C++, JavaScript, GO

AI/ML: Machine Learning, NLP, Computer Vision, Generative AI, Agentic AI, N8N Automation, RAG Systems

Frameworks: PyTorch, TensorFlow, Scikit-learn, Transformers, Diffusers, LangChain, LangGraph

Cybersecurity: SIEM (ELK Stack), Anomaly Detection, Threat Intelligence, Log Analysis

MLOps: Docker, Azure, AWS, FastAPI, Model Monitoring

Tools: Git, LaTeX, Linux, Jupyter, Weights & Biases, Vector Databases (Pinecone, ChromaDB)

Work Experience

AI Engineer **Jun 2025 – Aug 2025**
Ikonic, Islamabad, Pakistan

- Implemented multi-agent systems with LangChain, LangGraph, and Model Context Protocol (MCP) for agent communication
- Developed automation workflows using no-code tools to streamline repetitive tasks
- Engineered scalable architectures for agentic systems with focus on interoperability and modular integration

AI Engineer **May 2024 – Nov 2024**
Imarat Group of Companies, Islamabad, Pakistan

- Built AI-powered architectural design tool using diffusion models and ControlNet for floor plan generation and house rendering
- Leveraged Graph Neural Networks for spatial layout generation with style-based customization features

Deep Learning Engineer **Jan 2024 – Apr 2024**
Qlu.ai, Islamabad, Pakistan

- Designed RAG chat system using vector databases for data retrieval and transformer models for response generation
- Refactored AI service codebase to improve scalability and simplify integration with other systems
- Implemented model monitoring to track performance and reliability in production environments

Computer Vision Research Intern **Jun 2023 – Aug 2023**
DataInsight Lab, Islamabad, Pakistan

- Built multilingual handwritten text recognition pipelines for medical records using YOLO architecture
- Optimized model performance through preprocessing, data augmentation, and validation strategies

Selected Projects

ML-Enhanced Threat Intelligence Platform (Bachelor Thesis)

Technologies: Python, ELK Stack, PyTorch, Isolation Forest, Autoencoders

- Developed hybrid threat detection system combining unsupervised anomaly detection with rule-based techniques using real-world SIEM logs
- Engineered security features from enterprise log data, integrating real-time alerts with ELK Stack for incident response

Multi-Agent AI Trip Planning System

Technologies: Python, LangChain, LangGraph, MCP, FastAPI, PostgreSQL, Redis

- Built autonomous multi-agent system using Model Context Protocol for agent-to-agent and agent-to-tool communication
- Integrated flight, hotel, and Google Maps APIs with intelligent ranking based on reviews and user preferences

Intelligent Executive Interview Platform

Technologies: Python, OpenAI GPT-4, Whisper, Twilio, FastAPI

- Developed AI interview assistant generating contextual follow-up questions using NLP and real-time speech-to-text transcription
- Implemented structured agenda management for executive-level recruitment processes

RAG System for Email Query Processing

Technologies: Python, FastAPI, Custom-trained LLM, Pinecone, Redis

- Built Retrieval-Augmented Generation system for intelligent email data querying with semantic search and caching optimization
- Development halted due to security concerns regarding sensitive email data handling

Publications

In Preparation: Afzal, M. et al. (2025). *AI-Enhanced Architectural Design: From Interactive Floor Plan Generation to Style Rendering Using Diffusion Models and ControlNet.*

Certifications

- **Generative AI with LLMs** – DeepLearning.AI & AWS (2024)
- **Introduction to Generative AI** – DeepLearning.AI (2024)
- **Google Cybersecurity Professional Certificate** – Google (2024)
- **Microsoft Azure Fundamentals (AZ-900)** – Microsoft (2023)

Language Skills

- English (Fluent)
- German (Basic, improving)
- Urdu (Native)